

LECTURE NOTES ON AGT 227
BASIC FISHERIES TECHNOLOGY

1.0 DEVELOPMENT OF FISHERIES IN NIGERIA

1.1 HISTORY OF THE DEVELOPMENT OF THE FISHERIES INDUSTRY IN NIGERIA

Introduction: The fisheries industry in Nigeria has evolved from a purely subsistence-based activity to a significant economic sector contributing to food security, employment, and national income. Fish constitutes a major source of animal protein for Nigerians, and the sector supports millions of livelihoods across coastal, riverine, and inland communities.

The development of fisheries in Nigeria can be broadly divided into four major phases:

- (a) Pre-colonial era
- (b) Colonial era
- (c) Post-independence era
- (d) Modern/contemporary era

(a) Pre-Colonial Era (Before 1900)

Traditional Fisheries Practices

Before colonial influence, fishing was primarily Artisanal and subsistence-based; practiced by riverine and coastal communities such as the Ijaw, Ilaje, and Efik. Fishing was conducted using simple tools like hooks, traps, baskets, and canoes. Fishing activities were deeply rooted in culture and tradition, with knowledge passed orally across generations. There were no formal institutions or scientific management systems.

Socio-economic Importance:

- (i) Fish served as a major protein source.
- (ii) Surplus fish was preserved by smoking, drying, and salting.
- (iii) It contributed to local trade and barter systems.

Despite its importance, the industry remained undeveloped and unregulated.

(b) Colonial Era (1900–1960)

Early Institutional Development:

The foundation of modern fisheries in Nigeria began during colonial rule. In 1941, there was establishment of a Fisheries Development Branch under the colonial agricultural department. Fisheries surveys were conducted in Lagos and surrounding coastal waters; and early experiments in brackish water fish culture were carried out.

Formation of Fisheries Administration: From 1945 – 1950, fisheries administration was reorganized and later became the Federal Fisheries Service. Their focus was on resource assessment, improving fishing methods and introducing scientific approaches.

Introduction of Aquaculture: In 1951, there was the establishment of a large fish farm at Panyam (near Jos) for carp (*Cyprinus carpio*) culture. This marked the beginning of organized fish farming (aquaculture) in Nigeria. However, development during this period remained limited and primarily experimental.

(c) Post-Independence Era (1960–1980s)

Expansion and Policy Development: After independence in 1960, Nigeria began to recognize fisheries as an important economic sector and implemented National Development Plans targeting fish production. Between 1962 and 1985, several fisheries development programmes were introduced.

Establishment of Research Institutions: Key institutions were created to support fisheries development. In 1968, there was the establishment of the National Institute for Freshwater Fisheries Research (NIFFR) at Kainji, Niger State, Nigerian Institute for Oceanography and Marine Research (NIOMR), Lagos State and River Basin Development Authorities (RBDAs). These institutions focused on fish breeding and culture techniques, inland and marine fisheries research, and training of manpower.

Growth of Aquaculture: Aquaculture development occurred in two phases:

Phase 1: From 1950 – 1970, Introduction and awareness.

Phase 2: From 1970 – 1992 witnessed the expansion and establishment of fish farms.

Government and private sector participation increased, leading to construction of fish farms,

establishment of demonstration centres and increased interest in fish farming.

Oil Boom Impact: During the 1970s oil boom, massive importation of frozen fish reduced emphasis on local fish production. Aquaculture development received low priority.

(d) Modern Era (1990s – Present)

Structural Transformation: From the 1990s onward, fisheries in Nigeria experienced:

- (i) Increased private sector participation
- (ii) Growth of commercial aquaculture
- (iii) Introduction of improved technologies

Fish farming systems evolved into subsistence ponds, commercial farms and intensive systems such as recirculatory aquaculture systems (RAS).

Challenges in Fisheries Development: Despite progress, the industry faces several constraints:

- (i) Poor access to finance for artisanal fishers
- (ii) Inadequate infrastructure (cold storage, processing facilities)
- (iii) Overfishing and environmental degradation
- (iv) Pollution (especially in the Niger Delta)
- (v) Dependence on fish importation
- (vi) Weak enforcement of fisheries regulations

Contributions of Fisheries to Nigeria's Economy: The fisheries sector contributes significantly to:

- (i) Food security (major protein source)
- (ii) Employment generation
- (iii) Income generation
- (iv) GDP contribution and foreign exchange (through exports).

1.2 STATUS OF FISHERIES RESOURCES PRODUCTION IN THE NIGERIAN ECONOMY

Introduction: Fisheries resources production in Nigeria plays a crucial role in ensuring food security, employment generation, and economic development. The sector comprises capture fisheries (marine and inland) and culture fisheries (aquaculture/fish farming), both of which contribute significantly to national fish supply. Fish is one of the most affordable and accessible sources of animal protein in Nigeria, accounting for a substantial proportion of dietary intake among both rural and urban populations.

Overview of Fisheries Production in Nigeria

Current Production Levels: Nigeria's fisheries production has grown steadily over the decades. Recent data show that total fish production is approximately 1.0 – 1.4 million metric tonnes annually, domestic demand is about 3.3 million metric tonnes, leaving a deficit of over 2 million tonnes. This gap is largely filled through fish importation, making Nigeria one of the largest fish importers in Africa.

Growth Trends: Production increased from about 55,000 tonnes in 1961 to over 1 million tonnes today. The sector recorded peak production around 1.2 million tonnes in 2017. Recent government interventions have boosted production to about 1.4 million tonnes (2025 estimate).

Structure of Fisheries Production

(a) Artisanal Fisheries: This contributes over 70% of domestic fish production. They usually operate in coastal waters, rivers, lakes, and lagoons. Artisanal fisheries is characterized by small-scale operations, traditional fishing gears, high labor intensity. This subsector is the backbone of fish supply in Nigeria.

(b) Industrial Fisheries: This subsector contributes less than 5% of total production. It focuses on marine shrimp and fish trawling.

Challenges in this subsector include: high operational costs, illegal, unreported, and unregulated (IUU) fishing, declining fish stocks.

(c) Aquaculture (Fish Farming): Aquaculture is the fastest-growing subsector. The growth rate is about 6.7% annually. It contributes significantly to GDP growth compared to capture

fisheries. Major species cultured are Catfish (*Clarias gariepinus*) and Tilapia (*Tilapia spp*), *Oreochromis niloticus*. Aquaculture has evolved from subsistence ponds to commercial and intensive systems.

Contribution to the Nigerian Economy

(i) Contribution to GDP: Fisheries contribute approximately 1 – 5% to Nigeria's GDP. It plays a key role in the agricultural sector GDP.

(ii) Employment Generation: Over 1.4 million people are directly engaged in fisheries. Millions more are involved in fish processing, marketing and distribution, net making and boat building.

(iii) Food Security and Nutrition: Fish provides about 40% of animal protein intake in Nigeria. The average per capita consumption is about 13 kg/year.

(iv) Foreign Exchange and Trade: Nigeria spends significant foreign exchange on fish imports due to production deficits. The government is promoting import substitution policies to boost local production.

Resource Base of Fisheries in Nigeria

(a) Inland Water Resources: Nigeria has extensive inland water bodies: Rivers (e.g., Niger and Benue), Lakes (e.g., Kainji, Chad), Reservoirs and floodplains. These support inland fisheries, which contribute significantly to total production.

(b) Marine Resources: Nigeria has a coastline of about 853 km along the Atlantic Ocean. The marine resources is rich in finfish, crustaceans (shrimp, prawns), molluscs. However, marine fisheries are underexploited due to poor management and illegal fishing.

1.3 IMPORTANCE OF FISH IN HUMAN NUTRITION

Fish is highly valuable due to:

- (i) High-quality protein (essential amino acids)
- (ii) Rich in omega-3 fatty acids (brain development)
- (iii) Vitamins: A, D, B-complex
- (iv) Minerals: calcium, iodine, phosphorus

Examples include catfish (*Clarias gariepinus*) (widely consumed and cultured) and Tilapia (affordable protein source).

Major Fishing Areas in Nigeria

- (a) Inland Waters: Rivers, lakes, reservoirs. Example: Kainji Lake
- (b) Brackish Waters: Estuaries, lagoons. Example: Lagos Lagoon
- (c) Marine Waters: Coastal waters of the Atlantic Ocean

Major Water Bodies in Nigeria

- (a) Rivers. e.g. River Niger, River Benue
- (b) Lakes. e.g. Lake Chad, Kainji Lake
- (c) Flood Plains. e.g. Niger-Benue floodplains
- (d) Lagoons. e.g. Lagos Lagoon
- (e) Sea. e.g. Atlantic Ocean

1.4 HYDROLOGICAL CYCLE AND WATER BALANCE OF LAKES

Hydrological Cycle

Hydrology is simply the study of water. It is concerned with the origin, distribution, and properties of water on the globe. The water cycle (hydrologic cycle) is a biogeochemical cycle that involves the continuous movement of water on, above and below the surface of the Earth across different reservoirs. Hydrological cycle involves the following:

- (i) Evaporation
- (ii) Transpiration
- (iii) Condensation
- (iv) Precipitation
- (v) Runoff

(i) Evaporation: Evaporation is the change of state in a substance from a liquid to a gas. In meteorology, the substance we are concerned about the most is water. For evaporation to take place, energy is required. The energy can come from any source: the sun, the atmosphere, the Earth, or objects on the Earth such as humans.

(ii) Transpiration: Transpiration is the evaporation of water from plants through stomata. Stomata are small openings found on the underside of leaves that are connected to vascular plant tissues. In most plants, transpiration is a passive process largely controlled by the humidity of the atmosphere and the moisture content of the soil. Of the transpired water

passing through a plant only 1% is used in the growth process of the plant. The remaining 99% is passed into the atmosphere.

(iii) Condensation: Condensation is the process whereby water vapor is changed into a liquid state. In the atmosphere, condensation may appear as clouds or dew. This is also the process whereby water appears on the side of an uninsulated cold drink can or bottle. Condensation is not a matter of one particular temperature but of a difference between two temperatures: the air temperature and the dewpoint temperature. The dew point is the temperature at which dew can form - it is the point at which air becomes saturated and cannot hold any more water vapor. Any additional cooling causes water vapor to condense.

(iv) Precipitation: Precipitation results when tiny condensation particles, through collision and coalescence, grow too large for the rising air to support, and thus fall to the Earth. Precipitation can be in the form of rain, hail, snow, or sleet. Precipitation is the primary way we receive fresh water on Earth.

(v) Runoff: Runoff occurs when there is excessive precipitation and the ground is saturated (cannot absorb any more water). Rivers and lakes are results of runoff. Some runoff evaporates into the atmosphere, but most water in rivers and lakes returns to the oceans. If runoff water flows into a lake only with no outlet for water to flow out, then evaporation is the only means for water to return to the atmosphere. As water evaporates, impurities or salts are left behind. Evaporation of this runoff into the atmosphere begins the hydrologic cycle over again.

Water Balance Equation

Input = Output $\pm$ Storage

- Inputs: rainfall, inflow
- Outputs: evaporation, outflow

Role in Fish Production

- (i) Controls water level
- (ii) Influences nutrient availability

(iii) Affects breeding cycles

1.5 PROPERTIES OF AQUATIC ENVIRONMENT

(a) Physical Properties: Temperature, Light penetration, Turbidity

(b) Chemical Properties: Dissolved oxygen (DO), pH, Salinity

(c) Biological Properties: Plankton (primary productivity), Microorganisms

These factors determine fish survival, growth, and reproduction.

1.6 FISHERIES SUB-SECTORS IN NIGERIA

(a) Artisanal Fisheries

Types: (i) Subsistence and (ii) Small-scale commercial

Roles: Major source of local fish supply (ii) Employment generation.

Challenges/Problems: (i) Poor gear (ii) Post-harvest losses (iii) Lack of credit facilities.

(b) Industrial Fisheries

Roles: (i) Large-scale fish production (ii) Export earnings

Challenges/Problems: (i) High operational cost (ii) Illegal fishing (iii) Environmental degradation.

2.0 RELATIONSHIP BETWEEN HYDROGRAPHY AND FISHERIES

Hydrography: This is the study of the physical characteristics of water bodies. Hydrography directly influences fish production. Fisheries productivity depends on hydrographic conditions because fish survival, feeding, migration, and reproduction are controlled by water properties.

Key Relationships:

- Water temperature → affects metabolism and growth of fish
- Salinity → determines species distribution
- Currents → transport nutrients and larvae
- Depth & light penetration → influence primary productivity.

2.1 VARIOUS PARTS OF THE OCEAN

The ocean is divided based on depth and distance from shore:

Horizontal Zones:

- (i) Littoral zone – near shore
- (ii) Neritic zone – continental shelf (high productivity)
- (iii) Oceanic zone – open ocean

Vertical Zones:

- (i) Epipelagic (0–200 m) – sunlight zone (most fish)
- (ii) Mesopelagic (200–1000 m)
- (iii) Bathypelagic (>1000 m)

2.2 TYPES OF OCEAN CURRENTS

Ocean currents are large, continuous movements of seawater that flow through the world's oceans in specific directions.

Types of Currents

1. Based on Depth:

- (a) Surface Currents: This occurs in the upper layer of the ocean (about 0–400 m). It is mainly driven by wind and influenced by the Coriolis effect. It moves horizontally across large distances; e.g. Gulf Stream.
- (b) Deep Ocean Currents: It is found below the surface layer of the ocean. It is driven by differences in temperature and salinity (density). It forms part of the global system known as thermohaline circulation.

2. Based on Temperature:

- (a) Warm Currents: This current originates from the equator and move toward the poles. They carry warm water; e.g., Kuroshio Current.
- (b) Cold Currents: This current originates from polar or temperate regions and move toward the equator. They carry cold water; e.g., Canary Current.

3. Based on Cause (Driving Forces):

- (a) Wind-driven Currents: This is caused by prevailing winds. It is common in surface waters; e.g., Trade wind currents.

(b) Density-driven Currents: It is caused by differences in water density (temperature + salinity). It is also called thermohaline currents.

(c) Tidal Currents: It is caused by the gravitational pull of the Moon and Sun. It is associated with tides.

4. Based on Direction of Flow:

(a) Horizontal Currents: These current flows parallel to the ocean surface. It is the most common type.

(b) Vertical Currents: This current involves upward or downward movement of water. They include: Upwelling (brings nutrient-rich water to the surface) and Downwelling (sinks surface water).

2.3 EFFECTS OF OCEAN CURRENTS ON FISH DISTRIBUTION

(i) Carry nutrients → increase fish abundance

(ii) Influence migration patterns

(iii) Determine spawning grounds

2.4 UPWELLING

Upwelling is the upward movement of cold, nutrient-rich water from deep ocean layers to the surface.

Causes and Effects of Upwelling

Causes: (i) Wind action (ii) Earth's rotation (Coriolis effect) (iii) Ocean currents.

Effects on Fish Production:

(i) Increases plankton → more fish

(ii) Supports major fisheries worldwide

(iii) Enhances fish breeding areas

3.0 ESSENTIAL REQUIREMENTS FOR THE ESTABLISHMENT OF FISH FARMS

3.1 Definition and Scope of Aquaculture

Aquaculture is the rearing of aquatic organisms (such as fish, crustaceans, and plants) in a controlled/confined environment.

Scope Includes: (i) Fish farming (e.g., *Clarias gariepinus*) (ii) Hatchery operations (iii) Feed production (iv) Fish processing, preservation and marketing.

3.2 Pre-requisites for Establishing Aquaculture Enterprise

- (a) Suitable land (clay soil preferred for earthen ponds)
- (b) Reliable water supply
- (c) Capital investment
- (d) Technical knowledge
- (e) Access to market
- (f) Good infrastructure

3.3 Functions of Fish Ponds:

- (i) Provide controlled environment for fish growth (ii) Facilitate feeding and harvesting
- (iii) Enhance fish reproduction (iv) Protect fish from predators.

3.4 Classification of Fish Ponds

- (a) Based on Species: (i) Monoculture (one species) (ii) Polyculture (multiple species)
- (b) Based on Usage: (i) Nursery ponds, (ii) Rearing ponds (iii) Production ponds
- (c) Based on Water Source: (i) Natural water ponds (ii) Artificial ponds

3.5 Essential Cultural Requirements

- (i) Soil Type: Clay soil is best for pond construction because it retains water. Sandy soil is poor (leaks water). The soil should have at least 30% content.

- (ii) Water Quality Parameter

Parameter

Temperature

Dissolved Oxygen

pH

Turbidity

Salinity

4.0 CONSTRUCTION PROCEDURES OF A POND

4.1 Survey of Proposed Fish Farm Site

Site survey is the systematic assessment of a location to determine its suitability for fish farming.

Objectives:

- (i) Determine soil suitability
- (ii) Assess water availability and quality
- (iii) Evaluate topography
- (iv) Identify environmental risks

Factors to consider before siting a fish pond:

1. Soil Type: Clay or loamy soils are best (retain water). Avoid sandy or porous soils.
2. Water Supply: Reliable and adequate water source (e.g., streams, boreholes, rainfall) that is free from pollutants.
3. Topography: Gentle slope is preferred for easy drainage. Avoid flood-prone areas.
4. Accessibility: Nearness to roads and markets.
5. Climate: Suitable temperature range for fish growth.
6. Security: Protection from theft and predators

Survey Methods: (i) Physical inspection (ii) Soil testing (iii) Water analysis (iv) Use of leveling instruments.

4.2 BASIC OPERATIONS IN POND CONSTRUCTION

(a) Layout and Pegging:

- (i) Mark pond boundaries using pegs and ropes
- (ii) Ensure correct pond dimensions
- (iii) Establish water flow direction

(b) Excavation of Pond:

- (i) Removal of topsoil
- (i) Digging to required depth (1–1.5 m typical)
- (ii) Use manual or mechanical methods

(c) Shaping of Dykes/Laying out the Pond bottom:

(i) Construct embankments using excavated soil

(ii) Compact soil to prevent leakage

(iii) Maintain proper slope (usually 2:1 ratio)

(iv) Drainage Installation:

Sluice Gate/Monk: Controls water level and allows complete drainage

Inlet: This is the entry point for water and it should have a screen to prevent predators.

Outlet: Discharges water and prevents fish escape.

4.3 POND MANAGEMENT PRACTICES

(a) Pond Liming: Liming is the application of lime materials {e.g. agricultural lime (CaCO_3), quicklime (CaO), slaked lime; $\text{Ca}(\text{OH})_2$ } to pond water and /or bottom soil to improve water quality and soil conditions for fish production.

(b) Pond Fertilization: This is the application of organic (poultry droppings, cow dung, compost) or inorganic fertilizers (NPK fertilizers, urea) to increase the productivity of pond water for better fish growth.

(c) Feeding: This could be done twice daily or thrice depending on the stage (age) of the fish.

(d) Water quality monitoring: This involves measuring the physico-chemical parameters of water such as water temperature, water pH, Dissolved Oxygen (DO) etc.

(i) Aims and Actions of Liming

Aims: It neutralizes acidity, improve soil quality and enhance microbial activity.

Actions: Increases pH, releases nutrients and improves plankton growth.

ii) Aims and Actions of Fertilization

Aims: Increase natural fish food (plankton) and thus improve productivity

Actions: Supplies nutrients (nitrogen, phosphorus) and enhances primary productivity

General Rules for Using Fertilizers and Lime

(i) Apply lime before fertilization

- (ii) Avoid over-application
- (iii) Use based on soil test results
- (iv) Apply fertilizers in split doses

Types of Lime:

- (i) Agricultural lime (limestone/Calcium carbonate (CaCO_3))
- (ii) Quicklime (Calcium Oxide (CaO))
- (iii) Slaked lime (Calcium hydroxide - $\text{Ca}(\text{OH})_2$)

Actions: Raises pH and kills harmful organisms

4.4 WATER QUALITY PARAMETERS

- (i) **Water pH:** The water pH indicates acidity or alkalinity and it affects fish metabolism. It can be measured with the aid of a pH meter.
- (ii) **Water Temperature:** This is the degree of coldness or hotness of the water. It influences growth rate and affects oxygen solubility. It can be measured with the aid of a mercury in glass thermometer.
- (iii) **Dissolved Oxygen (DO):** This is the amount of oxygen that is available in water for aquatic life. It is essential for respiration and it determines fish survival. It can be measured with the aid of a Dissolved Oxygen (DO) meter or with the use of Winkler's solution.
- (iv) **Carbon Dioxide (CO_2):** This is a by-product of respiration and high levels are harmful. It is measured with a CO_2 meter.
- (v) **Water turbidity/transparency:** Water transparency refers to the clarity of water (how clear or turbid the water is). It is the degree to which light can penetrate through water. It is commonly measured using a Secchi disc. The depth at which the disc disappears from view is recorded as the transparency level. Water turbidity on the other hand refers to how muddy or unclear water appears due to the presence of fine particles or plankton. It is measured using a turbidity meter (in Nephelometric Turbidity Units – NTU).

4.5 EFFECTS AND IMPORTANCE OF WATER QUALITY PARAMETERS

- (a) **pH:** Ideal water pH range for fish culture is 6.5 – 8.5. Low pH causes stress, poor growth

while high pH causes toxicity.

(b) Temperature: The optimal water temperature range for most tropical aquaculture species is 25°C – 32°C. Below 20°C, growth slows significantly and above 35°C causes stress, reduced feeding, and possible mortality.

(c) Dissolved Oxygen: The optimal DO range for most cultured fish is 5 – 8 mg/L (ppm). Low DO causes fish suffocation while high DO leads healthy growth. (Note: ≥ 4 mg/L is the acceptable minimum for survival and moderate growth; stress begins and feeding reduces when the DO < 3 mg/L but < 2 mg/L is dangerous and can lead to mass mortality).

(d) Carbon Dioxide: Optimal CO₂ for fish culture is 0 – 5 mg/L. However, 5 – 10 mg/L is acceptable but may begin to affect fish. High CO₂ reduces oxygen uptake and causes fish stress

5.0 CULTURE OF BRACKISH AND FRESHWATER FISH

5.1 Management of Fish Ponds/Farm

Fish pond management involves all practices required to ensure optimal fish growth, health, and productivity in a culture system.

Key Management Practices

1. Pond Preparation:

- (i) Draining and drying
- (iii) Liming (to adjust pH)
- (ii) Removal of predators and weeds
- (iv) Fertilization (organic/inorganic)

2. Water Quality Management:

- (i) Temperature: 25 – 32°C (ideal for tropical fish)
- (iii) pH: 6.5 – 8.5

3. Feeding Management:

- (i) Use of balanced feed
- (ii) Dissolved oxygen: >5 mg/L

- (iv) Regular water exchange
- (ii) Feeding rate: 3–5% body weight (fingerlings)
- (iii) Feeding frequency: 2–3 times daily

4. Stock Monitoring:

- (i) Sampling for growth
- (ii) Mortality check

5. Predator and Pest Control:

- (i) Birds, snakes, frogs
- (iii) Adjustment of feeding rate
- (ii) Use of netting/scare devices

6. Record Keeping:

- (i) Stocking data
- (iii) Growth performance
- (ii) Feed usage
- (iv) Mortality

5.2 CULTURE OF BRACKISH AND FRESHWATER FISH SPECIES

Examples: Tilapia (*Tilapia* spp.) and Heterotis (*Heterotis niloticus*)

(i) Distribution/Occurrence

Tilapia: it's widely distributed in Africa and are found in rivers, lakes, reservoirs.

Heterotis niloticus: This species is native to West and Central Africa. They are found in slow-moving rivers and floodplains.

(ii) Selection of Species for Culture

Criteria: (i) Fast growth rate (ii) Disease resistance (iii) Market demand (iv) Availability of fingerlings (v) Feed conversion efficiency (FCE): this refers to how efficiently a fish converts feed to flesh, low FCE is desirable

(iii) Spawning and Growth of Fingerlings

Tilapia: They are mouth brooders, early maturity (3–6 months) and have high fecundity.

Heterotis: They build nests, parental care present and they have moderate growth rate.

(iv) Problem of Overpopulation: This is common in tilapia due to uncontrolled breeding. It leads to competition for food and stunted growth.

Control Measures: These include mono-sex culture (male tilapia), use of predators, regular harvesting.

(v) Stocking System

Types: (1) Monoculture: Stocking one species of fish in a pond

(2) Polyculture: Stocking more than one species in a pond

(3) Integrated fish farming: combining fish farming with other branches of agriculture like poultry, rice farming, vegetable farming, etc.

Stocking Density: This depends on the system of aquaculture being practiced

5.3 FISH FEED PRODUCTION TECHNIQUES

Steps in Feed Production: (1) Ingredient selection (2) Grinding (3) Mixing (4) Pelleting (5) Drying (6) Storage.

Flow chart showing feed production: raw materials → grinding → mixing → pelleting → drying → packaging.

5.4 TYPES OF FISH FEED AND FEEDSTUFF

Types of Feed:

(a) Natural Feed: Plankton, algae

(b) Artificial Feed: Pellets (floating/sinking)

(c) Supplementary Feed: Maize bran, rice bran

Feedstuff Categories:

(i) Protein sources: fish meal, soybean meal

(ii) Energy sources: maize, cassava

(iii) Vitamins & minerals

(iv) Lipids (fats/oils)

5.5 IDENTIFICATION OF DISEASED FISH

Physical Signs: Lesions or ulcers, discoloration, fin rot, swollen abdomen.

Behavioural Signs: Erratic swimming, gasping at surface, loss of appetite, isolation from group.

Common Diseases:

- (a) Bacterial infections
- (b) Fungal infections
- (c) Parasitic infestations

6.0 HARVESTING PROCESS OF FISH IN PONDS

6.1 Common Fishing Equipment Used in Nigeria

Fishing equipment refers to tools and materials used in the capture of fish in inland and marine waters.

Categories of Fishing Equipment

A. Nets

- (i) Gill nets – trap fish by gills
- (ii) Cast nets – circular nets thrown by hand
- (iii) Seine nets – used to encircle fish
- (iv) Lift nets – raised vertically

B. Traps and Pots: They are made from bamboo, cane, or wire mesh. Examples are basket traps, drum traps.

C. Hooks and Lines: These include single hook, longline, handline. A bait is used to attract fish.

D. Spears and Harpoons: Used in shallow waters, they are common in artisanal fisheries.

E. Accessories: These include floats, sinkers, ropes, swivels.

6.2 Harvesting fish by draining pond water

This method is common in aquaculture systems.

Procedure:

1. Stop feeding fish 24 hours before harvest
2. Lower water level gradually
3. Open drainage outlet (monk or sluice gate)
4. Allow water to flow out slowly
5. Collect fish at the drainage point
6. Use scoop nets to gather remaining fish

Advantages

- (i) Complete harvest
- (ii) Easy sorting and counting

Disadvantages

- (i) Stressful to fish
- (ii) Time-consuming

6.3 HARVESTING FISH WITHOUT DRAINING WATER

- a. Seine Netting: This is a large net dragged across pond. It requires 2–5 people.
- b. Cast Net: They are thrown manually; suitable for small-scale harvesting.
- c. Scoop Net: This is a hand-held net, used for selective harvesting.
- d. Gill Net: Fish get entangled by gills
- e. Traps (Drum, Basket, Dyke traps): Fish enter but cannot escape
- f. Hook and Line: Used for large or predatory fish
- g. Fillet Net (Gill Net Type): Mesh size determines fish caught.

6.4 FISHING CRAFTS

Fishing crafts are vessels used in fishing operations.

Types:

- (i) Raft – simple floating structure made of logs
- (ii) Canoe – dugout or planked wood
- (iii) Dinghy – small boat often used as auxiliary craft
- (iv) Motorized boats – fitted with engines

(v) Trawlers – large commercial fishing vessels

6.5 Materials Used in Boat/Craft Construction

- a. Wood: Common in Nigeria and they are cheap and easy to repair.
- b. Steel: They are strong and durable; they are used for large vessels.
- c. Fiberglass (Glass Fibre): They are lightweight and are resistant to corrosion.
- d. Aluminium: Light and rust-resistant
- e. Plastic: Used for small crafts.

6.6 DIFFERENCE BETWEEN OUTBOARD AND INBOARD MOTORS

Feature	Outboard Motor	Inboard Motor
Location	Outside boat	Inside boat
Portability	Removable	Fixed
Maintenance	Easier	More complex
Power	Lower	Higher
Cost	Cheaper	Expensive

7.0 FISH PRESERVATION, PROCESSING AND DISTRIBUTION METHODS

7.1 Importance of Proper Fish Handling and Preservation

Fish is highly perishable due to:

- High moisture content (70 – 80%)
- High protein (supports microbial growth)
- Enzymatic activity after death

Importance of Proper Fish Handling:

1. Prevents spoilage
2. Maintains quality (taste, texture, nutrients)
3. Reduces post-harvest losses (very high in Nigeria)
4. Improves food safety
5. Extends shelf life
6. Enhances market value and export potential
7. Ensures consumer health

Stages of Fish Spoilage:

(a) Rigor mortis: This is the stiffening of fish muscles after death due to biochemical changes in the body.

(b) Autolysis (self-digestion): This is the breakdown of cells and tissues by the organism's own enzymes after death.

(c) Bacterial decomposition: this is the breakdown of dead organic matter by microorganisms (especially bacterial), resulting in spoilage and decay.

7.2 FISH PRESERVATION METHODS

(a) Chilling: This is a process in which fish is stored in ice at temperatures 0 – 4°C.

Advantages: (i) Slows microbial growth (ii) Maintains fresh quality

Disadvantages: (i) Short-term preservation (few days) (ii) Expensive to run (iii) Relies on stable electricity

(b) Freezing: This is a process in which fish is stored in a refrigerator at temperature -18°C or lower.

Advantages: (i) Long shelf life (months)

Disadvantages: (i) Expensive equipment (ii) Requires constant power supply

(c) Smoking: This involves the use of heat and smoke from wood for preservation.

Types: (i) Hot smoking (ii) Cold smoking

Functions: (i) Dries fish, (ii) Adds flavor, (iii) Preserves

Example (equipment): Traditional kiln, Improved kiln (e.g., Chorkor smoker)

(d) Salting: This is the application of dry salt or brine for preservation. Salting removes moisture (osmosis) and inhibits microbial growth.

Types: (i) Dry salting (ii) Wet salting (brining)

(e) Canning: Fish sealed in airtight containers and heat sterilized. Examples: Sardines, tuna.

Advantages: (i) Very long shelf life (1–5 years) (ii) Safe from contamination.

Disadvantages: (i) Expensive to run (ii) Requires high technology

7.3 DISTRIBUTION AND MARKETING CHANNELS IN NIGERIA

Fish Distribution Chain: (1) Fishermen/Fish farmers (2) Middlemen (wholesalers) (3) Processors (4) Retailers (5) Consumers

Channels: (i) Direct sales (farm gate) (ii) Local markets (e.g., Makoko Fish Market) (iii) Cold chain distribution (iv) Export markets.

Challenges: (i) Poor transportation (ii) Lack of cold storage (iii) Price fluctuations (iv) Post harvest losses.

Improvement Strategies: (i) Cooperative societies (ii) Cold chain development (iii) Government policies (iv) Improved roads and infrastructure.

7.4 ROLE OF PACKAGING AND STORAGE IN FISHERIES

Packaging Functions:

1. Protection from contamination
2. Prevent moisture loss/gain
3. Improve shelf life
4. Ease transportation
5. Branding and marketing

Packaging Materials: Plastic bags, vacuum packs, cans, cartons.

Storage Methods:

- (i) Cold storage (refrigerators, cold rooms)
- (ii) Dry storage (for smoked/salted fish)

Importance: (i) Maintains quality (ii) Prevents spoilage (iii) Reduces losses

7.5 TOOLS AND EQUIPMENT USED IN FISH PROCESSING AND PRESERVATION

Common Equipment

1. Smoking kiln: Traditional kiln, Chorkor smoker
2. Cold room: Large-scale storage facility
3. Refrigerator: Small-scale cold storage
4. Freezer: Used for deep freezing
5. Drying racks: Used for air drying

- 6. Knives and cutting tools: Used for fish dressing
- 7. Weighing scales: Used for measurement
- 8. Ice boxes: Used for transportation of fish.

1. Fish Smoking Kiln (Chorkor Kiln)

Parts: (i) Smoking chamber – where fish is arranged on trays

(ii) Wire mesh trays – hold fish during smoking

(ii) Firewood chamber (hearth) – heat and smoke source

(iii) Chimney/vent – allows smoke circulation and exit

(iv) Mud or metal walls – retain heat

(v) Access door – for loading/unloading fish

Function: (a) Dries fish (b) Infuses smoke flavor (c) Preserves fish by reducing moisture

2. Cold Room (Fish Storage Facility)

Parts: (i) Insulated walls – prevent heat

(ii) Evaporator fan unit – circulates cold air

(i) Compressor – compresses refrigerant

(ii) Condenser – releases heat outside

(iii) Shelving/racks – store fish

(iv) Door with rubber seal – prevents cold loss

Function: (a) Maintains temperature (0°C to -18°C) (b) Preserves fish for long periods

3. Freezing System (Refrigeration Cycle)

Main Components:

(i) Compressor – increases pressure of refrigerant

(ii) Condenser coil – releases heat

(iii) Expansion valve – reduces pressure

(iv) Evaporator coil – absorbs heat (cooling effect)

4. Packaging Types in Fisheries

Common Packaging Types:

- (i) Plastic bags (polyethylene) – fresh/chilled fish
- (ii) Vacuum packaging – removes air, extends shelf life
- (iii) Cans (metal containers) – for canned fish
- (iv) Cartons/boxes – transport and bulk storage

Functions:

- (i) Protect from contamination
- (ii) Extend shelf life
- (iii) Improve marketability
- (iv) Ease transportation

ASSIGNMENT

1. Write on any ten adaptive features of fish that make it able to live in water
2. Draw a map of Nigeria showing the major water bodies.

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